

assignment_kine1D_1

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:51pm

Quiz Instructions

2 attempts

tutorial in dropbox



Question 1 6.7 pts

Which of the following quantities has units of a displacement? (There could be more than one correct choice.)

☐

-120 m/s

☐

32 ft/s² vertically downward

☐

186,000 mi

☐

9.8 m/s²

☐

40 km southwest



Question 2 6.7 pts

A motorist travels 160 km at 80 km/h and 160 km at 100 km/h. What is the average speed of the motorist for this trip?

☐

90 km/h

☐

91 km/h

☐

89 km/h

☐

84 km/h



Question 3 6.7 pts

If you run a complete loop around an outdoor track of length 400 m in 100 s, find your

(a) average velocity and (b) average speed.

☐

(a) + 4m/s (b) 4 m/s

☐

(a) - 400 m/s (b) 400 m/s

☐

(a) 0 (b) 4 m/s



Question 4 6.7 pts

You are driving home on a weekend from school at 55 mi/h for 110 miles. It then starts to snow and you slow to 35 mi/h. You arrive home after driving 4 hours and 15 minutes. How far is your hometown from school?

☐

200 mi

☐

180 mi

☐

190 mi

☐

210 mi



Question 5 6.7 pts

A mosquito flying at 3 m/s that encounters a breeze blowing at 3 m/s in the opposite direction has a speed of

☐

0 m/s.

☐

3 m/s.

☐

4 m/s.

☐

6 m/s.



Question 6 6.7 pts

A race car circles 10 times around a circular 8.0-km track in 20 min. Using SI units

(a) what is its average speed for the ten laps?

(b) what is its average velocity for the ten laps?

☐

(a) 67m/s (b) 0

☐

(a) 240 km/h (b) 240 km/h



(a) 4 km/h (b) 0



(a) 67m/s (b) + 67m/s



Question 7 6.7 pts

The average speed of a horse that gallops 10 kilometers in 30 minutes is



15 km/h.



40 km/h.



30 km/h.



20 km/h.



Question 8 6.7 pts

A polar bear starts at the North Pole. It travels 1.0 km south, then 1.0 km east, and then 1.0 km north to return to its starting point. This trip takes 45 min. What was the bear's average velocity?



0.067 km/h



5.3 km/h



4.0 km/h



0.00 km/h



Question 9 6.7 pts

A light-year is the distance that light travels in one year. The speed of light is 3.00×10^8 m/s. How many miles are there in one light-year? (1 mi = 1609 m, 1 y = 365 d)

 9.46×10^{12} mi 5.88×10^{12} mi 5.88×10^{15} mi 9.46×10^{15} mi

Question 10 6.7 pts

The position $x(t)$ of a particle as a function of time t is given by the equation $x(t) = (3.5 \text{ m/s})t - (5.0 \text{ m/s}^2)t^2$. What is the average velocity of the particle between $t = 0.30 \text{ s}$ and $t = 0.40 \text{ s}$?

☐

0 m/s

☐

-2 m/s

☐

3 m/s

☐

2 m/s



Question 11 6.7 pts

If you are driving 72 km/h along a straight road and you look to the side for 4.0 s, how far do you travel during this inattentive period?

☐

18 m

☐

80 m

☐

20 m

☐

40 m



Question 12 6.7 pts

A runner ran the marathon (approximately 42.0 km) in 2 hours and 57 min. What was the average speed of the runner in m/s?

☐

3.95 m/s

☐

14.2 m/s

☐

14,200 m/s

☐

124 m/s



Question 13 6.7 pts

What must be your average speed in order to travel 350 km in 5.15 h?

☐

66.0 km/h

☐

69.0 km/h

☐

67.0 km/h

☐

68.0 km/h



Question 14 6.7 pts

Which of the following quantities has units of a velocity? (There could be more than one correct choice.)

☐

40 km southwest

☐

9.8 m/s downward

☐

-120 m/s

☐9.8 m/s² downward☐

186,000 mi



Question 15 6.7 pts

An object moves 15.0 m north and then 11.0 m south. Find both the distance it has traveled and the magnitude of its displacement.

☐

4.0 m, 26.0 m

☐

26.0 m, 26.0 m

☐

26.0 m, 4.0 m

☐

4.0 m, 4.0 m



Question 16 2 pts

bonus

Acceleration is 2m/s/s. time elapsed is 4s. initial velocity is 0. what is the displacement.

A) **8 m**

B) **16 m**

C) **32 m**

D) **64 m**

☐

16m

☐

32m

☐

4m

☐

64m

Not saved

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